



ComS 207: Programming I

Instructor: Alexander Stoytchev

http://www.cs.iastate.edu/~alex/classes/2006_Spring_207/



Introduction and First Program

January 9, 2006

*ComS 207: Programming I (in Java)
Iowa State University, SPRING 2006
Instructor: Alexander Stoytchev*



Outline

- **Questionnaire**
- **Class Overview**
- **Syllabus**
- **First Program**

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Questionnaire

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A silly question

- **How many of you know how to read and write?**

Well, this was not always the case

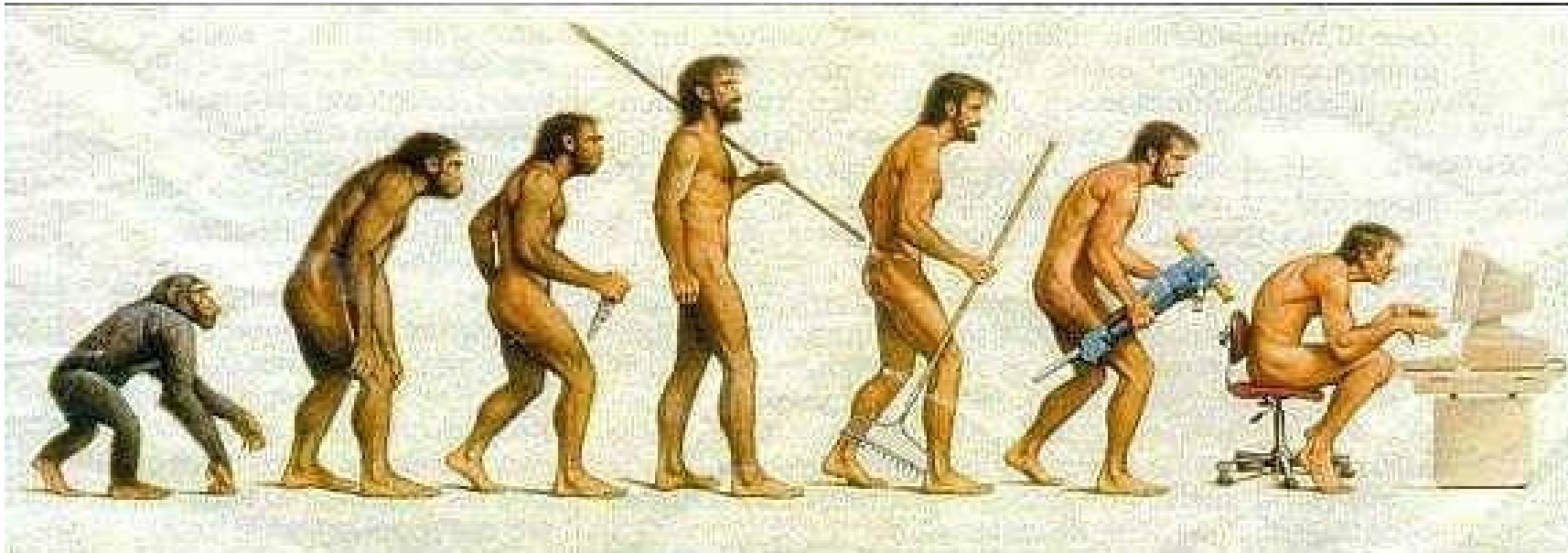


A silly question

program

- How many of you know how to ~~read and write?~~

Today programming is becoming a required skill for many jobs



Language Acquisition





[<http://www.wellroundedkids.com/store/Accessories/wooden%20alphabet%20puzzle.jpg>]

Alphabet Manuel LSF



Dessin : Albert TABAOT (Sourd)

[<http://www.usm67.org/alphabet/images/alphabet.jpg>]



Java

- **A *programming language* specifies the words and symbols that we can use to write a program**
- **A programming language employs a set of rules that dictate how the words and symbols can be put together to form valid *program statements***
- **The Java programming language was created by Sun Microsystems, Inc.**
- **It was introduced in 1995 and it's popularity has grown quickly since**

Reserved Words

- The Java reserved words:

<code>abstract</code>	<code>else</code>	<code>int</code>	<code>strictfp</code>
<code>boolean</code>	<code>enum</code>	<code>interface</code>	<code>super</code>
<code>break</code>	<code>extends</code>	<code>long</code>	<code>switch</code>
<code>byte</code>	<code>false</code>	<code>native</code>	<code>synchronized</code>
<code>case</code>	<code>final</code>	<code>new</code>	<code>this</code>
<code>catch</code>	<code>finally</code>	<code>null</code>	<code>throw</code>
<code>char</code>	<code>float</code>	<code>package</code>	<code>throws</code>
<code>class</code>	<code>for</code>	<code>private</code>	<code>transient</code>
<code>const</code>	<code>goto</code>	<code>protected</code>	<code>true</code>
<code>continue</code>	<code>if</code>	<code>public</code>	<code>try</code>
<code>default</code>	<code>implements</code>	<code>return</code>	<code>void</code>
<code>do</code>	<code>import</code>	<code>short</code>	<code>volatile</code>
<code>double</code>	<code>instanceof</code>	<code>static</code>	<code>while</code>



Focus of the Course

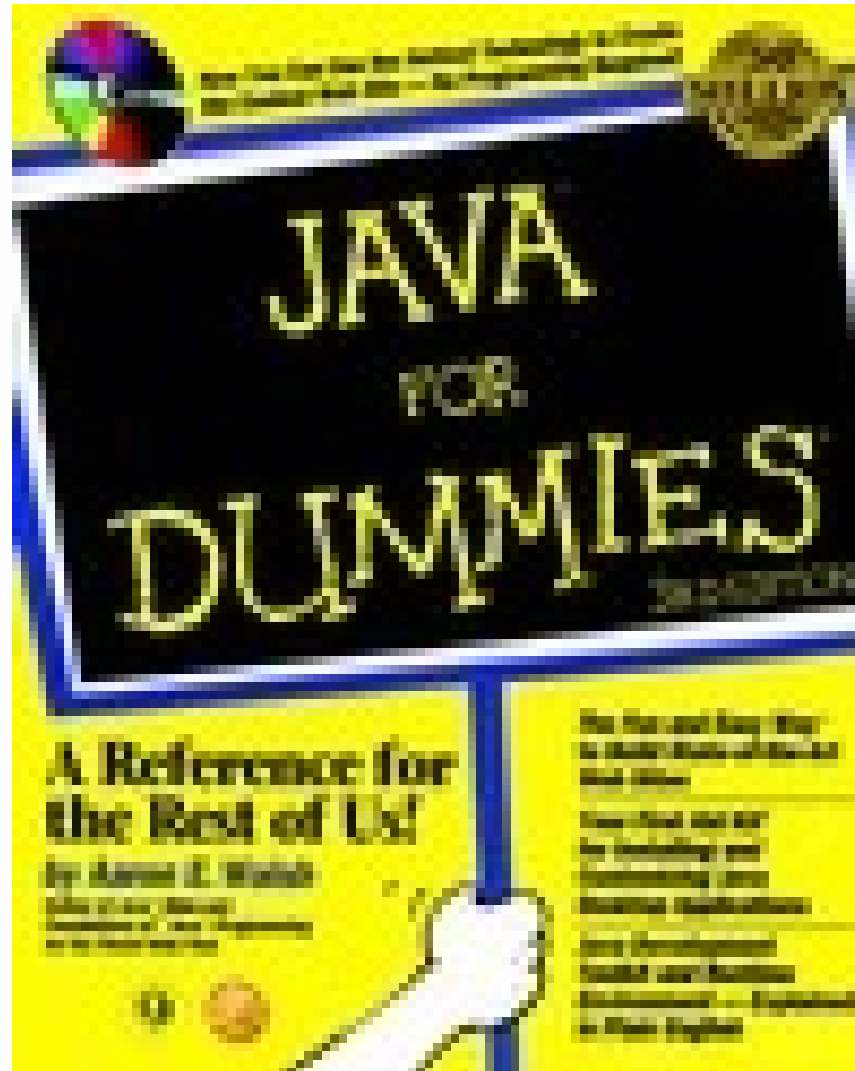
- **This is an intro to programming class
(that uses the java programming language)**
- **The focus is on**
 - **the logic of programming**
 - **problem solving**
 - **program design, implementation, and testing**
 - **object-oriented concepts**
 - **classes**
 - **Objects**
 - **methods**



Focus of the Course

- **This class is geared for non-Computer Science majors.**
- **If your major is CS you should take ComS 208.**
- **However, this does not mean that this is an easy class...**

In other words, this class is NOT





What this class will NOT cover

- **We will not go into details of programming GUIs**
- **We will not cover many of the libraries and classes that ship with the java language**
- **Currently there are > 2,500 classes in these libraries**
- **We will not cover some advanced Object-Oriented concepts**



Warning

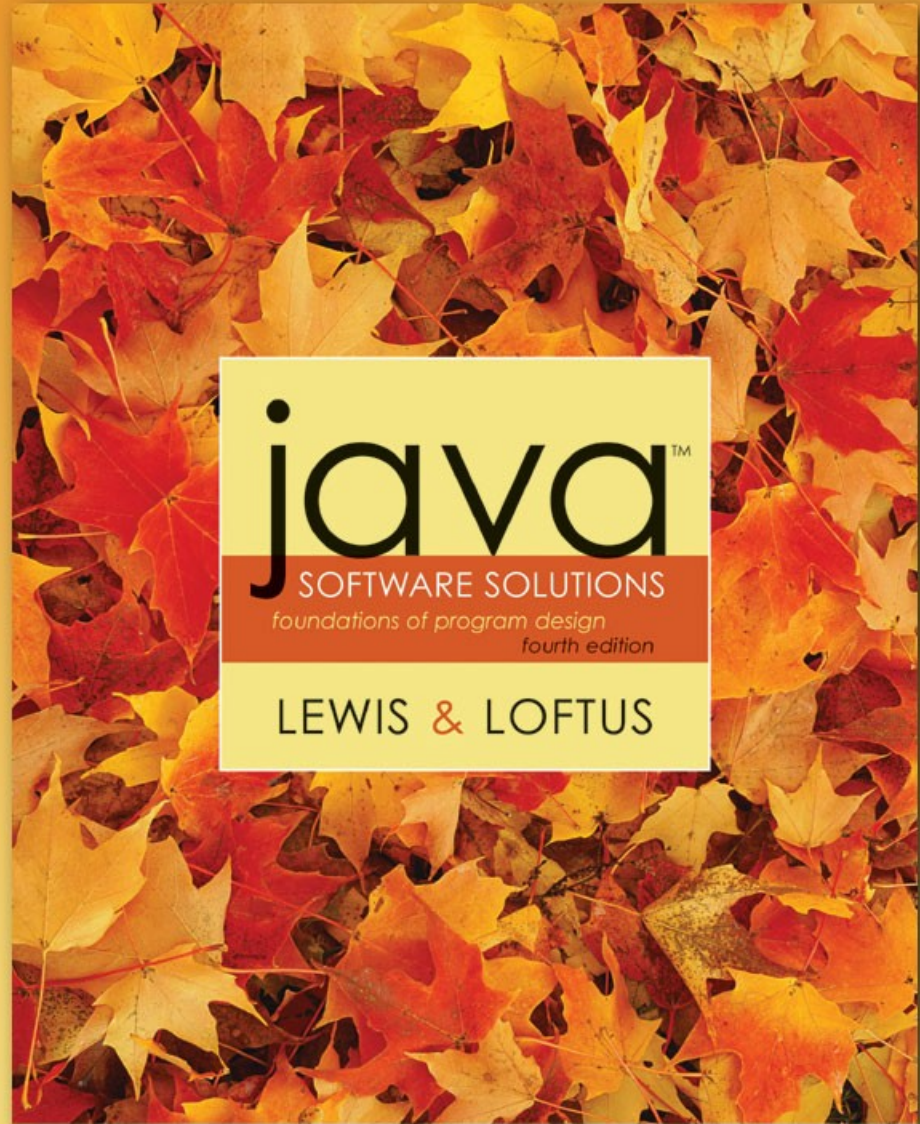
- **Learning how to program takes a lot of time!**
- **It also requires a lot of patience.**
- **You cannot learn how to program by just reading a textbook. You have to spend long hours in front of the computer.**
- **If you want to learn how to program well you will have to take at least 2-3 classes. This class alone is not enough.**



Syllabus

- **Posted on the Class Web Page:**
 - http://www.cs.iastate.edu/~alex/classes/2006_Spring_207/

REQUIRED TEXTBOOK



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First Program

Java Program Structure

- In the Java programming language:
 - A program is made up of one or more *classes*
 - A class contains one or more *methods*
 - A method contains program *statements*
- These terms will be explored in detail throughout the course
- A Java application always contains a method called `main`

Java Program Structure

```
//  comments about the class
public class MyProgram
{
    //  comments about the method
    public static void main (String[] args)
    {
        System.out.println("Hello World!");
    }
}
```

Java Program Structure

```
// comments about the class
```

```
public class MyProgram
```

```
{
```

class header

class body

Comments can be placed almost anywhere

```
}
```

Java Program Structure

```
// comments about the class
```

```
public class MyProgram  
{
```

```
    // comments about the method
```

```
    public static void main (String[] args)
```

```
    {
```

```
    }
```

```
}
```



method body



method header

Java Program Structure

```
//  comments about the class
public class MyProgram
{
    //  comments about the method
    public static void main (String[] args)
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        System.out.println("Hello World!");
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Questions/Comments?



THE END